

A Residual Set of Norm-Weak USC Points for M -Embedded Spaces

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Source Question

In *Computing sub differential limits of operators on Banach spaces*, arXiv:2212.09650, Rao studies norm-weak upper semicontinuity of the preduality map. After recalling the state-space criterion of Godefroy–Indumathi and proving results for M -ideals, the paper asks:

We do not know how to determine the largeness of points of norm-weak usc for the preduality map on X^{**} in the category sense, when X is a M -ideal in its bidual.

Result

The category answer is positive for every non-reflexive M -embedded Banach space.

Theorem 1. *Let X be a non-reflexive Banach space which is an M -ideal in its bidual X^{**} . Then the set of points of norm-weak upper semicontinuity for the preduality map on X^{**} contains a dense G_δ subset of X^{**} . In particular, the norm-weak usc points form a residual subset of X^{**} .*

Idea of the Proof

Rao’s Proposition 9(b) already proves norm-weak usc at every $\tau \in X^{**}$ satisfying two conditions:

$$\tau \text{ attains its norm on } X^* \quad \text{and} \quad \text{dist}(\tau, X) < \|\tau\|.$$

The missing category observation is that, for M -embedded X , these two conditions hold simultaneously on a residual set. Indeed X^* has the Radon–Nikodym property, so Bourgain’s theorem, or equivalently the Stegall variational principle in its standard norm-attainment form, gives a residual set of norm-attaining functionals in $(X^*)^* = X^{**}$. The strict inequality $\text{dist}(\tau, X) < \|\tau\|$ defines a dense open subset of X^{**} . Their intersection is dense G_δ , and Rao’s proposition applies pointwise on that intersection.

Proof

Proof. We work over the real scalars, as in the source paper.

Let

$$\mathcal{U} = \{\tau \in X^{**} : \text{dist}(\tau, X) < \|\tau\|\}.$$

The set $\mathcal{U}$ is open, because both $\tau \mapsto \text{dist}(\tau, X)$ and $\tau \mapsto \|\tau\|$ are norm-continuous.

Lemma 1. *$\mathcal{U}$ is dense in X^{**} .*

Proof. Fix $\tau \in X^{**}$ and $\varepsilon > 0$. If $\text{dist}(\tau, X) < \|\tau\|$, there is nothing to prove. Suppose instead that $\text{dist}(\tau, X) = \|\tau\| = M$.

If $M = 0$, choose any nonzero $x \in X$ with $\|x\| < \varepsilon$. Then $\tau + x = x \in \mathcal{U}$.

Assume $M > 0$. Pick $0 < r < \varepsilon$. Choose $x^* \in S_{X^*}$ such that $|\tau(x^*)| > M - r/3$. Then choose $y \in S_X$ with $|x^*(y)| > 2/3$, and replace y by $-y$ if needed so that $\tau(x^*) + rx^*(y)$ has modulus greater than M . Put $x = ry$. Since $x \in X$, passage to the quotient by X gives

$$\text{dist}(\tau + x, X) = \text{dist}(\tau, X) = M.$$

On the other hand,

$$\|\tau + x\| \geq |(\tau + x)(x^*)| > M.$$

Thus $\tau + x \in \mathcal{U}$ and $\|\tau + x - \tau\| = r < \varepsilon$. Hence $\mathcal{U}$ is dense. $\square$

Since X is M -embedded, X^* has the Radon–Nikodym property. By Bourgain’s theorem on RNP sets, in the form recalled by Jung–Martin–Rueda Zoca, the absolutely strongly exposing scalar functionals on X^* form a residual subset of $(X^*)^* = X^{**}$. Every such functional attains its norm on X^* . Therefore

$$\mathcal{N} = \{\tau \in X^{**} : \tau \text{ attains its norm on } X^*\}$$

contains a dense G_δ subset of X^{**} .

Consequently $\mathcal{R} = \mathcal{U} \cap \mathcal{N}$ contains a dense G_δ subset of X^{**} . For every $\tau \in \mathcal{R}$, Rao’s Proposition 9(b) applies directly: τ attains its norm on X^* and $\text{dist}(\tau, X) < \|\tau\|$, hence τ is a point of norm-weak usc for the preduality map on X^{**} . Thus the norm-weak usc points contain a dense G_δ subset. Their complement is contained in the complement of this dense G_δ , hence is meagre. $\square$

Concrete Corollary

Taking $X = c_0(\Gamma)$, for any infinite set Γ , gives $X^{**} = \ell_\infty(\Gamma)$. Thus the norm-weak usc points for the preduality map on $\ell_\infty(\Gamma)$ are residual. In this discrete family there is also an explicit dense open subfamily: bounded functions whose supremum norm is attained at one coordinate with a strict gap from all other coordinates. The earlier c_0 -only packet in this folder’s history used this concrete route; the theorem above subsumes it.

Verification and Limitations

The proof uses three standard ingredients:

1. Rao’s Proposition 9(b) from the source paper.
2. The standard structural fact that if X is M -embedded, then X^* has the Radon–Nikodym property.
3. Bourgain’s RNP norm-attainment theorem, equivalently the Stegall variational principle in the form that absolutely strongly exposing scalar functionals on an RNP space form a residual subset of the dual.

The theorem addresses the source paper’s category question for non-reflexive M -embedded Banach spaces. It does not address the separate operator-space intermediate question in Remark 13 of the source paper.

Novelty Check

Local index search for arXiv:2212.09650, norm-weak usc, preduality map, and M -embedded category terms found only the earlier c_0 partial packet from this run. Web searches for exact phrases such as norm-weak usc preduality map M -embedded residual, preduality map M -ideal in its bidual category, and norm-weak upper semi-continuity c_0 preduality map did not locate an explicit statement of the full category answer. The proof is nevertheless a short corollary of Rao's Proposition 9(b) and Bourgain–Stegall residual norm-attainment.

Source Evidence

when is $\lim_{t \rightarrow 0+} \frac{\|x+t\tau\|-\|x\|}{t} = \sup\{\tau(f) : f \in S_x\}$ for all $\tau \in X^{**}$?
We know that $\lim_{t \rightarrow 0+} \frac{\|x+t\tau\|-\|x\|}{t} = \sup\{\tau(f) : f \in S_x\}$ for all $\tau \in X^{**}$.

Thus we need equality of the suprema for all $\tau \in X^{**}$. This is clearly seen to be equivalent to S_x being weak*-dense in S^x .

We next recall that for $x^* \in X^*$ that attains its norm at a $x \in X_1$, the map $\rho(x^*) = S_x$ is called the preduality map. We note that points of X^* that attain the norm, is norm dense in X^* .

Definition 1. We recall from [4] that x^* is said to be a point of norm weak usc for ρ , if given a weak neighbourhood V of 0 in X , there is a $\delta > 0$ such that for a norm attaining x_1^* with $\|x^* - x_1^*\| < \delta$ implies $\rho(x_1^*) \subset V + \rho(x^*)$.

Thus by Lemma 2.2 in [4] the denseness of the state space is equivalent to x being a point of norm-weak upper-semi-continuity of the preduality map $\rho(x) = S_x$ on X^{**} . More generally, the same conclusions apply for any point of norm-weak usc in a dual space X^* (it is assumed that the functional attains its norm).

Since in many classical situations, like in ℓ^p spaces ($1 < p < \infty$) or when X, Y are reflexive with the metric approximation property, one has the inclusion map on the space of compact operators, $\mathcal{K}(X, Y) \subset \mathcal{L}(X, Y)$ is the canonical embedding of $\mathcal{K}(X, Y)$ in its bidual $\mathcal{L}(X, Y)$ (see the discussion on page 247 in [2]), we take the help of norm-weak usc operators in $\mathcal{K}(X, Y)$ or $\mathcal{L}(X, Y)$ to evaluate the limits. We refer to the monograph [2], for basic theory of tensor products and structure of spaces of operators.

We show that when the projective tensor product space $X \otimes_\pi Y$ has the Radon-Nikodym property (RNP), then for $A \in \mathcal{L}(X, Y^*)$, $\lim_{t \rightarrow 0+} \frac{\|A+tB\|-\|A\|}{t} = \sup\{x^{**}(B^*(y^*)) : x^{**} \otimes y^* \in S_A\}$, when A is a point of norm-weak usc for the preduality map on $\mathcal{L}(X, Y^*) = (X \otimes_\pi Y)^*$.

canonical projection Q on X^{***} defined above is a L -projection. See Chapter III of [5] for examples and Chapter VI when spaces of compact operators $\mathcal{K}(X, Y)$ is a M -ideal, when $\mathcal{L}(X^{**}, Y^{**})$ is identified as its bidual. When X is a M -ideal in its bidual, X^* has the RNP. See [5] Theorem III.3.1.

It is easy to see that unitaries in a C^* -algebra A , remain as points of norm-weak usc for the preduality map in all duals of higher even order of A (as they are still unitaries in the bigger space). The following phenomenon exhibits weak*-dense set of points of norm-weak usc for the preduality map on X^{**} . We do not know how to determine the largeness of points of norm-weak usc for the preduality map on X^{**} in the category sense, when X is a M -ideal in its bidual.

Proposition 9. *Let X be a non-reflexive space that is a M -ideal in its bidual. a) Any $0 \neq x \in X$ is a point of norm-weak usc for the preduality map on X^{**} and it continues to be so in all higher duals of even order of X .*

*b) If $\tau \in X^{**}$ is such that τ attains its norm on X^* and $d(\tau, X) < \|\tau\|$, then τ is a point of norm-weak usc for the preduality map on X^{**} . This behaviour continues in all the duals of even order of X .*

Proof. (a): Since the projection Q is a L -projection, $X^{***} = X^* \oplus_1 X^\perp$, we see that x^{***} is the unique extension of its restriction to X and the kernel annihilates X . Hence the conclusion follows.

It was shown in [8] that under the hypothesis X continues to be a M -ideal of all the higher order even duals of X (all in canonical embeddings). Hence the conclusion follows from Proposition 6.

(b): Since $X^{***} = X^* \oplus_1 X^\perp$ for the canonical projection, the conclusion follows from arguments similar to the ones given during the proof of Theorem 5.

Human Review Recommendation

Review as a claimed full solution to the category-largeness question after Rao's Proposition 9. The main checks are:

1. confirm that the cited Bourgain–Stegall residual theorem applies to the Banach space X^* ;
2. confirm that the set $\mathcal{U} = \{\tau : \text{dist}(\tau, X) < \|\tau\|\}$ is dense in X^{**} ;
3. confirm that Rao's Proposition 9(b) has no additional hidden hypothesis beyond non-reflexive M -embeddedness, norm attainment on X^* , and the strict distance inequality.

References

1. T. S. S. R. K. Rao, *Computing sub differential limits of operators on Banach spaces*, arXiv:2212.09650, 2022.
2. G. Godefroy and V. Indumathi, *Norm-to-weak upper semi-continuity of the duality and pre-duality mappings*, Set-Valued Analysis 10 (2002), 317–330.
3. P. Harmand, D. Werner, and W. Werner, *M-ideals in Banach spaces and Banach algebras*, Lecture Notes in Mathematics 1547, Springer, 1993.
4. M. Jung, M. Martin, and A. Rueda Zoca, *Residuality in the set of norm attaining operators between Banach spaces*, J. Funct. Anal. 284 (2023), Paper No. 109746; arXiv:2203.04023.